(1) Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2) $(D^2+4D+4)y = \cos x$

(3) $(D^2-4D-5)y = e^{3x} + 4\cos 3x$

(4) $(D^2+9)y = \cos 3x$

(5) $(D^2+1)y = \sin x \sin 2x$ (6) $(2D^2-5D+3)y = \cos 3x \cos 2x$

Case III when $\phi(x) = x^m$, m being a +ve integer

Here we have to evaluate $\frac{1}{f(D)} x^m$. For that expand $\frac{1}{f(D)} = [f(D)]^{-1}$ in ascending powers of D and operate on x^m. We use the following results for expansion.

$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots$ (1)

$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$ (2)

$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots$ (3)

$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \cdots$ (4)

Solve the following diff. eqs.

(1) $(D^2+2D+1)y = 2x + x^2$

(2) $(D^2+D+1)y = x^2$

(3) $(D^2+25)y = x^2 + x$

Answers

(1) $(D^2+2D+1)y = 2x + x^2$

CF is got by solving $(D^2+2D+1)y = 0$

A. E. is $D^2+2D+1 = 0 \Rightarrow (D+1)^2 = 0$

$\Rightarrow (D+1)(D+1) = 0 \Rightarrow D = -1, -1$

$CF = (c_1 + c_2 x) e^{-x}$

$PI = \frac{1}{D^2+2D+1}(2x+x^2) = \frac{1}{(D+1)^2}(2x+x^2)$

$= (1+D)^{-2}(2x+x^2)$ using series (4)

$= (1 - 2D + 3D^2 - 4D^3 + \cdots)(2x+x^2)$

$\therefore \quad PI = (2x+x^2) - 2D(2x+x^2) + 3D^2(2x+x^2) - \cdots$

$= (2x+x^2) - 2(2+2x) + 3(2) + 0$

$= x^2 + 2x - 4 - 4x + 6$

$= x^2 - 2x + 2$

Gen. soln. $y = CF + PI$

$= (c_1 + c_2 x)e^{-x} + x^2 - 2x + 2$

(2) $(D^2+D+1)y = x^2$

CF is got by solving $(D^2+D+1)y = 0$

A.E is $D^2+D+1 = 0$

Solving $D = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2}$ $\quad \alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2}$

$= \frac{-1 \pm i\sqrt{3}}{2}$

$CF = e^{-\frac{x}{2}}\left[c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x\right]$

$PI = \frac{1}{D^2+D+1}(x^2)$

$= (1+D+D^2)^{-1}(x^2)$

$= x^2 - (D+D^2)(x^2) + (D+D^2)^2(x^2) - \cdots$

$= x^2 - (2x+2) + (D^2+2D^3+D^4)(x^2) - \cdots$

$= x^2 - (2x+2) + (2+0+0) - 0$

$= x^2 - 2x - 2 + 2$

$= x^2 - 2x$

Gen. solution: $y = e^{-\frac{x}{2}}\left[c_1 \cos\frac{\sqrt{3}x}{2} + c_2 \sin\frac{\sqrt{3}x}{2}\right] + x^2 - 2x$

(3) $(D^2+25)y = x^2 + x$

CF is got by solving $(D^2+25)y = 0$

A.E. is $D^2+25 = 0 \Rightarrow D^2 = -25 \Rightarrow D = \pm 5i$

$CF = c_1 \cos 5x + c_2 \sin 5x$

$PI = \frac{1}{D^2+25}(x^2+x) = \frac{1}{25\left(\frac{D^2}{25}+1\right)}(x^2+x)$

$= \frac{1}{25}\left(1+\frac{D^2}{25}\right)^{-1}(x^2+x)$

$$PI = \frac{1}{25}\left(1 - \frac{D^2}{25} + \frac{D^4}{625} - \cdots\right](x^2+x)$$

$$= \frac{1}{25}\left[(x^2+x) - \frac{D^2}{25}(x^2+x) + \frac{D^4}{625}(x^2+x) + \cdots\right]$$

$$= \frac{1}{25}\left[x^2+x-\frac{2}{25}+0\right] = \frac{1}{625}(25x^2+25x-2)$$

Gen. soln: $y = c_1\cos 5x + c_2 \sin 5x + \frac{1}{625}(25x^2+25x-2)$

Solve (1) $(D^2-2D)y = 2+3x$

(2) $(D^2-1)y = 3+2x^2$ (3) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2 + e^{4x}$

(4) $(D^2+4D+5)y = e^{2x} + \cos x + x^2$

Case IV when $\phi(x) = e^{ax}\cdot V$ where V is a function of $\sin ax$, $\cos ax$ or x^m

$$D(e^{ax}\cdot V) = e^{ax}D(V) + ae^{ax}V$$
$$= e^{ax}(D+a)V$$

$$D^2(e^{ax}\cdot V) = D\left[e^{ax}(D+a)V\right]$$
$$= e^{ax}\cdot D(D+a)V + (D+a)V(ae^{ax})$$
$$= e^{ax}\left[D(D+a) + a(D+a)\right]V$$
$$= e^{ax}\left[(D+a)(D+a)\right]V$$
$$= e^{ax}(D+a)^2 V$$

$$\therefore f(D)(e^{ax}\cdot V) = e^{ax} f(D+a)V$$

This shows that $\frac{1}{f(D)}(e^{ax}\cdot V) = e^{ax}\cdot\frac{1}{f(D+a)}V$

Problems Solve $(D^2+3D+2)y = e^{2x}\sin x$

To find CF solve $(D^2+3D+2)y = 0$

A.E. is $D^2+3D+2 = 0$ $\therefore D = -1, -2$

$CF = c_1 e^{-x} + c_2 e^{-2x}$

$$P.I = \frac{1}{D^2+3D+2}(e^{2x}\sin x)$$

$$= e^{2x}\cdot\frac{1}{(D+2)^2+3(D+2)+2}(\sin x)$$ (Replace D by D+2)

$$= e^{2x}\frac{1}{D^2+4D+4+3D+6+2}(\sin x)$$

$$= e^{2x}\cdot\frac{1}{D^2+7D+12}(\sin x)$$

$$= e^{2x}\frac{1}{-1^2+7D+12}(\sin x)$$ ($a=1$ Replace D^2 by -1^2)

$$= e^{2x}\cdot\frac{1}{7D+11}(\sin x)$$

$$= e^{2x}\cdot\frac{7D-11}{49D^2-121}(\sin x)$$ (Replacing D^2 by -1^2)

$$= e^{2x}\frac{7D-11}{49(-1^2)-121}(\sin x)$$

$$= e^{2x}\frac{7D-11}{-170}(\sin x)$$

$$= e^{2x}\left(\frac{-1}{170}\right)(7\cos x-11\sin x)$$

$$= \frac{1}{170}e^{2x}(11\sin x-7\cos x)$$

gen. soln. $y = CF + PI$

$$= c_1e^{-x}+c_2e^{-2x}+\frac{1}{170}e^{2x}(11\sin x-7\cos x)$$

(2) Solve $(D^2-2D+2)y = e^x x^3$

To find CF solve $(D^2-2D+2)y=0$

The auxillary eq is $D^2-2D+2=0$

$$D = \frac{2\pm\sqrt{4-8}}{2} = \frac{2\pm 2i}{2} = 1\pm i$$

$$CF = e^x(c_1\cos x + c_2\sin x)$$

$$PI = \frac{1}{D^2-2D+2}(e^x.x^3) = e^x\cdot\frac{1}{(D+1)^2-2(D+1)+2}(x^3)$$

$$= e^{x} \frac{1}{D^2+2D+1-2D-2+2} (x^3)$$

$$= e^{x} \frac{1}{D^2+1} (x^3)$$

$$= e^{x} (1+D^2)^{-1} (x^3)$$

$$= e^{x} [1 - D^2 + D^4 - D^6 + \dots] (x^3)$$

$$= e^{x} (x^3 - 6x + 0) = e^{x} (x^3 - 6x)$$

$\therefore$ gen. soln

$$y = CF + PI = e^{x} [c_1 \cos x + c_2 \sin x] + e^{x} (x^3 - 6x)$$

(3) Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^{-2x} \sin x$

CF is got by solving $(D^2+5D+6)y = 0$

A.E. is given by $D^2+5D+6=0$

ie $(D+2)(D+3) = 0 \Rightarrow D = -2, -3$

$$CF = c_1 e^{-2x} + c_2 e^{-3x}$$

$$PI = \frac{1}{D^2+5D+6} (e^{-2x} \sin x)$$

$$= e^{-2x} \cdot \frac{1}{(D-2)^2 + 5(D-2) + 6} (\sin x)$$

$$= e^{-2x} \frac{1}{D^2 - 4D + 4 + 5D - 10 + 6} (\sin x)$$

$$= e^{-2x} \frac{1}{D^2+D} (\sin x)$$

$$= e^{-2x} \cdot \frac{1}{-1^2 + D} (\sin x) \quad \left(\begin{matrix} a = 1 \\ \text{Replacing } D^2 \text{ by } -1^2 \end{matrix} \right)$$

$$= e^{-2x} \cdot \frac{1}{D-1} (\sin x)$$

$$= e^{-2x} \cdot \frac{D+1}{D^2-1} (\sin x) = e^{-2x} \cdot \frac{(D+1)}{-2} (\sin x)$$

$$= \frac{e^{-2x}}{-2} (\cos x + \sin x)$$

Gen. soln : $y = CF + PI$

i.e. $y = c_1 e^{-2x} + c_2 e^{-3x} - \frac{e^{-2x}}{2}(\sin x + \cos x)$

(4) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x e^x \sin x$

CF is got by solving $(D^2 - 2D + 1)y = 0$

A.E is $D^2 - 2D + 1 = 0 \Rightarrow D = 1, 1$

$CF = (C_1 + C_2 x) e^x$.

$PI = \frac{1}{D^2 - 2D + 1}(x e^x \sin x)$

$= \frac{1}{(D-1)^2}[e^x (x \sin x)]$

$= e^x \cdot \frac{1}{(D+1-1)^2}(x \sin x)$

$= e^x \cdot \frac{1}{D^2}(x \sin x)$ $\left(\frac{1}{D} \equiv \int\right)$

$= e^x \cdot \frac{1}{D} \int x \sin x$

$= e^x \frac{1}{D}[x(-\cos x) - 1(-\sin x)]$ (Applying Integration by Parts)

$= e^x \int(-x \cos x + \sin x)dx$ $\left[\int uv\,dx = uv_1 - u'v_2\right]$

$= e^x \left[\int -x \cos x\,dx + \int \sin x\,dx\right]$

$= e^x[-x \sin x + 1(-\cos x) - \cos x]$

$= e^x[-x \sin x - 2\cos x]$

$= e^x[-x \sin x + 2\cos x]$

$= -e^x[x \sin x + 2\cos x]$

Gen. soln: $y = (C_1 + C_2 x) e^x - e^x[x \sin x + 2\cos x]$

$= e^x[C_1 + C_2 x - (x \sin x + 2\cos x)]$